A same-disorder covariant channel in the Slater representation

Mixed words, a new cyclic direction and a stationary tracial sector

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Abstract

We add a microscopic covariant size descendant of the Hamiltonian to the selected Slater algebra. A marked-site extension of the normalized CAR defect expansion gives all fixed mixed words and compact-time convergence in the same Gaussian disorder. The new vector has a strictly positive component outside the entire previous cyclic space. Its joint transfer still has a compact selected operator, but a long-time state now retains a noncommutative two-color q-Gaussian sector. We identify the GNS representation of this particular stationary state, not the full quotient algebra or every stationary state. The results concern the specified annealed double-scaled model; the gravitational dictionary remains to be constructed.

1 Model and the constructive question

The previous selected algebra contained only $h, M, M^\dagger$. Its stationary states killed the compact selected channel and retained only a commutative energy algebra. We now enlarge the microscopic observable set before taking the limit. The fixed-charge chord framework is established in [1]; q-Gaussian Wick calculus in [2]; the dependence of the representation on the chosen observable algebra is central to [3, 5]. Here the additional operator is a descendant of the original random Hamiltonian, with its mixed correlations derived in the selected state.

Use standard CAR, even N, $m = N/2$, a fixed occupied set O and its empty complement E. For ordered p-element subsets write

$$H_N = \sqrt{w_N} \sum_{A,B} J_{AB} C_A^\dagger C_B, \qquad \mathbb{E} J_{AB} J_{CD} = \delta_{AD}\delta_{BC}. \tag{1}$$

The matrix J is Hermitian Gaussian; $A \cap B$ is allowed. Set

$$L_N = \binom{m}{p}^2, \qquad D_N = \binom{m}{p}\binom{m+p}{p}, \qquad h_N = H_N/\sqrt{w_N D_N}, \tag{2}$$

$$M_N^\dagger = L_N^{-1/2} \sum_{A\subset E,\, B\subset O} J_{AB} C_A^\dagger C_B, \qquad \varphi_N(X) = \mathbb{E}\langle O|X|O\rangle. \tag{3}$$

Both subset sizes in the last sum are p. Throughout, $\lambda_N = p^2/N \to \lambda > 0$, $q = e^{-4\lambda}$, $\gamma = q^{1/4}$ and $s = \sqrt{q}$. Time is conjugate to dimensionless h; the physical string energy is $(J_s/\sqrt{\lambda})h$, with $\hbar = 1$.

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2 The microscopic descendant and marked local sources

Let $\mathcal{L}$ denote full-Fock half-Clifford-size, acting on operators before restriction to the central charge sector. Our new field is

$$u_N = \frac{(p - \mathcal{L} - \lambda_N)h_N}{\sqrt{\lambda_N}}. \tag{4}$$

It is Hermitian, linear in the same J, and $U(N)$-equivariant. It is the normalized first size fluctuation used in the earlier bilocal expansion, not an independent matter ensemble.

For $X = C_A^\dagger C_B$, an own-overlap site in $A \cap B$ contributes a number operator n. At that site $p - \mathcal{L}$ replaces n by $I/2$; one sums over all own-overlap sites. Equivalently, its exponential replaces n by $n + (e^z - 1)I/2$. Give each differentiated endpoint a commuting nilpotent source z_i, $z_i^2 = 0$. The local replacement is then exactly $n + z_i I/2$. Single-membership sites have no source.

Retain the fixed empty/occupied Slater CAR factors and the Jordan signs from the previous site polynomial. For an unrestricted Gaussian edge with endpoints i, j, its own two-label collision, summed over empty and occupied sites, has polynomial

$$1 + \tfrac{1}{2}z_i + \tfrac{1}{2}z_j + \tfrac{1}{2}z_i z_j. \tag{5}$$

The constant term comes from the occupied site. The two linear terms each give $\langle n/2 \rangle$; the quadratic term gives $1/4$ on both occupations. A selected edge forbids this own overlap. Double membership belonging to two different edges has no source: neither endpoint has both labels of its own edge at that site. Higher memberships will be controlled after normalization, rather than discarded by an unnormalized rarity argument.

Centering the endpoint by $-\lambda_N$ multiplies its source by $e^{-\lambda_N z_i}$. The activity of a double cluster tends to $m(p/m)^2 = 2\lambda$. Thus the source-dependent exponent from (5) is $\lambda(z_i + z_j) + \lambda z_i z_j$. After centering, an unrestricted edge has source $e^{\lambda z_i z_j}$. A selected edge with a differentiated unrestricted endpoint has instead $e^{-\lambda z_i}$. Dividing each derivative by $\sqrt{\lambda}$ gives the edge coefficients

$$w(H,H) = w(U,U) = 1, \quad w(H,U) = 0, \quad w(H,P) = \gamma, \quad w(U,P) = -\sqrt{\lambda}\,\gamma. \tag{6}$$

Here P denotes the selected endpoint of a compatible edge; an edge with two selected endpoints has weight one. Its orientation restriction is unchanged. The derivation of the U–selected coefficient is especially important: tracial orthogonality of h and u does not imply orthogonality to $M^\dagger$ in the selected state.

3 Normalized defect proof of the all-word limit

Fix a word of length $2k$ and one Gaussian pairing. An edge is unrestricted (G) if neither endpoint is selected; otherwise its label is restricted by the selected endpoint(s). For each occupation $o \in \{0,1\}$, let $n_{o,S}$ count sites with membership subset S of the $2k$ edge labels, $|S| \geq 2$. Define the deficits $d_i = \sum_{o,S \ni i} n_{o,S}$, so the remaining singleton count is $p - d_i$. The exact normalized coefficient, divided by L_N^k, has terms

$$\frac{(m)_{V_0}(m)_{V_1}}{(m)_p^{2k}} \prod_i (p)_{d_i} \prod_{o,S} \frac{f_o(S;\boldsymbol{z})^{n_{o,S}}}{n_{o,S}!}. \tag{7}$$

Here $(x)_r$ is falling factorial and V_o is the number of used sites of occupation o: the number of correct singletons plus the number of defect sites. The source polynomial f_o is the local CAR matrix element with its crossing sign. As in the unmarked proof, wrong singletons vanish.

Equip the finite nilpotent coefficient ring with the sum of absolute coefficients. It is submultiplicative, and all local polynomials have norm at most a fixed B_k, independent of N, p. The singleton polynomials are unchanged. The factorial estimate in (7) therefore gives, uniformly on a compact positive λ_N range,

$$\|\text{term}\| \leq C_k \prod_{o,S} \frac{(B_k \alpha_S)^{n_{o,S}}}{n_{o,S}!}, \qquad \alpha_S = m(p/m)^{|S|}. \tag{8}$$

There are finitely many S. Their total activity is bounded, while $\sum_{|S|\geq 3} \alpha_S = O_k(p^3/N^2) \to 0$. The exponential majorant has a uniformly summable tail in total defect number. For a backward selected edge at least one required singleton count is zero, forcing $d_i = p$ and at least p incident defects; the same tail makes its normalized contribution vanish. This also controls configurations containing unrestricted edges and sources.

For a fixed double-defect configuration the background factor is $e^{-2\lambda k(k-1)}$. In the unmarked sum the double coefficients give $k_G + k(k-1) - 2\nu$, where k_G counts unrestricted edges and ν counts crossings plus nestings whose inner edge is selected. Consequently the scalar contribution divided by L_N^k is $e^{2\lambda k_G} q^\nu$. Since $D_N/L_N \to e^{2\lambda}$, normalization of the unrestricted endpoints supplies γ for each mixed edge. Equation (5) supplies the extra source factors already computed, without changing crossings or nestings. Dominated summation in the coefficient norm justifies differentiation of the limit.

Proposition 1. *Every fixed word in $h_N, u_N, M_N, M_N^\dagger$ converges in φ_N. Its limit is the sum over compatible all-forward pairings of the edge coefficients (6), times q^ν. Odd words vanish.*

All-forward means that the right creation endpoint of each selected edge acts first. The proposition is an all-order fixed-word result; the finite sixth-order checks below are independent diagnostics.

4 Actual finite-time convergence and the rank interface

The statement above concerns microscopic operators, and can be extended to genuine exponentials of h_N. For real independent Gaussian coordinates write $h_N = \sum_a x_a A_a$, $u_N = \sum_a x_a U_a$ and $M_N = \sum_a x_a B_a$. Covariance and irreducibility of $\bigwedge^m \mathbb{C}^N$ give

$$\sum_a A_a^\dagger A_a = \sum_a A_a A_a^\dagger = I, \qquad \sum_a U_a^2 = v_N I, \qquad v_N = \varphi_N(u_N^2) \to 1. \tag{9}$$

Hermiticity of the U_a follows from (4). For the selected block the two coefficient sums have eigenvalues $\binom{x}{p}^2/L_N$ and $\binom{m-x}{p}^2/L_N$, respectively, with $0 \le x \le m$; both are bounded by one.

In each Wick contraction open a Gaussian label with its column map, close it with its row map and retain the remaining labels as noise slots. Crossings permute slots and do not change their norms. A paired contraction is therefore bounded by the product of the coefficient row/column bounds. Since v_N is bounded, words with a fixed number of U insertions have the same factorial envelope as in the previous time-convergence argument, up to a constant depending on those insertions. With $F_d = \sqrt{(2d-1)!!}$, a total exponential degree n and fixed inserted degree d, the vector bound is CF_{n+d}. For T the sum of absolute complex times, the Taylor remainder is bounded by

$$C \sum_{n \ge R} \frac{T^n F_{n+d}}{n!} \le \frac{C T^R F_{R+d}}{R!\,(1-\rho_R)}, \qquad \rho_R = \frac{T\sqrt{2(R+d)+1}}{R+1} < 1, \tag{10}$$

once R is large enough that the ratios decrease. Fixed-word convergence followed by this uniform tail proves convergence on every compact complex-time window. No quenched concentration claim is made.

There is also an actual tensor-rank version of this field. The prior central-sector rank/size estimate, in the normalized annealed Hilbert–Schmidt norm, is

$$\|(\mathcal{R} - \mathcal{L})h_N\|_{2,m} = O(N^{-1}). \tag{11}$$

This difference is again a linear Gaussian equivariant Hermitian field. Its covariance is therefore $O(N^{-2})I_m$, so its row/column norm is $O(N^{-1})$. Expanding a difference of fixed mixed words and applying the same Wick-slot estimate proves that one may replace u_N everywhere above by

$$\widetilde{u}_N = (p - \mathcal{R} - \lambda_N)h_N/\sqrt{\lambda_N}. \tag{12}$$

The compact-time statement follows from the same tails. This step uses linear Gaussian covariance, not merely a tracial small-norm assertion. It does not yet include exact pinching of the neutral slope: pinching depends nonlinearly on J, and full-trace estimates alone do not control selected excited states.

5 Three-letter transfer and its Hilbert-space meaning

Use words of length n in H, U, P, ordered oldest to newest. Creation $a_H^\dagger, a_U^\dagger, b^\dagger$ appends the indicated letter. Set $c_H = \gamma$, $c_U = -\sqrt{\lambda}\gamma$. On a word $x_1 \cdots x_n$, a_i deletes each ordinary letter of matching color at position j with weight q^{n-j}, deletes each P with weight $c_i q^{n-1}$, and gives zero on an unmatched ordinary color. The operator b deletes each H, U, P with respective weights $c_H, c_U, 1$, all multiplied by q^{n-1}. Then

$$h = a_H + a_H^\dagger, \qquad u = a_U + a_U^\dagger, \qquad M = b. \tag{13}$$

The Gram form is defined by reverse-bra closure, or equivalently by cutting the pairing diagrams at the intermediate level. Positivity is inherited from the microscopic limiting moments. Distinct levels are orthogonal; quotient null vectors before completion. These strings are not a Euclidean orthonormal basis. Each level has dimension at most 3^n; no generic equality is assumed.

Direct deletion gives

$$a_i a_j^\dagger - q a_j^\dagger a_i = \delta_{ij} I, \tag{14}$$

$$a_i b^\dagger - q b^\dagger a_i = b a_i^\dagger - q a_i^\dagger b = c_i q^{\mathcal{N}}, \tag{15}$$

$$b b^\dagger - q b^\dagger b = q^{\mathcal{N}}. \tag{16}$$

With $[n]_q = (1-q^n)/(1-q)$, the first relation inductively gives $\left\| a_i^\dagger |_{\mathcal{H}_n} \right\|^2 \le [n+1]_q$. The last gives $\left\| b^\dagger |_{\mathcal{H}_n} \right\|^2 \le (n+1) q^n$. Thus h, u are bounded and b is compact: its tails tend to zero and all levels are finite.

The a_H primitive subspaces generate orthogonal q-Jacobi towers. Starting with $a_H \eta = 0$, the normalized vectors $(a_H^\dagger)^n \eta / \sqrt{[n]_q!}$ have Jacobi coefficients $\sqrt{[n+1]_q}$. Their complete orthogonal decomposition follows level by level by separating the range of $a_H^\dagger$ and its orthogonal complement. Hence h is a direct sum of standard q-Hermite Jacobi operators and is purely absolutely continuous.

Equation (16) places $q^{\mathcal{N}}$ in $\mathcal{A}_+ = C^*(1, h, u, M)$. Its powers converge in norm to the vacuum projection. Cyclicity then generates every compact operator: $\mathcal{K}(\mathcal{H}) \subset \mathcal{A}_+$ and $\mathcal{A}_+'' = B(\mathcal{H})$. Writing $\mathcal{D} = C^*(1, h, u)$, one obtains

$$\mathcal{A}_+ = \mathcal{D} + \mathcal{K}(\mathcal{H}). \tag{17}$$

The sum is closed because the image of $\mathcal{D}$ under the quotient map by $\mathcal{K}(\mathcal{H})$ is a closed C^*-algebra. Compactness of M has therefore been proved anew in the enlarged representation, rather than inferred from compactness on the smaller cyclic subspace.

6 The new direction and four concrete mixed tests

Let $\mathcal{H}_{hM}$ be the closed cyclic subspace generated by $h, M, M^\dagger$. Its level-n part is spanned by H/P strings: this follows inductively by subtracting the known annihilation pieces from h and using $M^\dagger$ for P. Since $u\Omega$ has level one, projecting onto the entire old cyclic space requires only the span of $h\Omega, M^\dagger\Omega$. Their Gram matrix and overlaps are

$$G_1 = \begin{pmatrix} 1 & \gamma \\ \gamma & 1 \end{pmatrix}, \qquad \begin{pmatrix} \langle h\Omega, u\Omega\rangle \\ \langle M^\dagger\Omega, u\Omega\rangle \end{pmatrix} = \begin{pmatrix} 0 \\ -\sqrt{\lambda}\gamma \end{pmatrix}. \tag{18}$$

Consequently

$$P_{hM}u\Omega = \frac{\sqrt{\lambda}s}{1-s}h\Omega - \frac{\sqrt{\lambda}\gamma}{1-s}M^\dagger\Omega, \tag{19}$$

$$r_\lambda := \|(1-P_{hM})u\Omega\|^2 = 1 - \frac{\lambda s}{1-s} = 1 - \frac{\lambda}{e^{2\lambda}-1} > 0. \tag{20}$$

In fact $1/2 < r_\lambda < 1$ for $\lambda > 0$ and $r_\lambda \to 1/2$ as $\lambda \downarrow 0$. This last statement is a limit of the derived formula, not an interchange with the microscopic double-scaling limit. The residual $\eta = (1-P_{hM})u\Omega$ satisfies $a_H\eta = 0$, because $a_H u\Omega = 0$, $a_H h\Omega = \Omega$ and $a_H M^\dagger\Omega = \gamma\Omega$. It starts a new Hamiltonian tower.

For comparison with future bilocal insertions, the all-word rule gives

$$\varphi(Mu^2M^\dagger) = 1 + \lambda s(1+q), \tag{21}$$

$$\varphi(uM^\dagger Mu) = \lambda s, \tag{22}$$

$$\varphi(MuM^\dagger u) = q(1+\lambda s), \tag{23}$$

$$\varphi(MM^\dagger u^2) = 1. \tag{24}$$

The two-point data are $\varphi(hu) = 0$, $\varphi(u^2) = 1$ and $\varphi(uM^\dagger) = -\sqrt{\lambda}\gamma$. They distinguish this same-disorder descendant from a freely assigned independent probe. The nonzero residual is also stronger than a low-level numerical rank increase: level orthogonality establishes its orthogonality to every vector of the old cyclic space.

There is an exact finite-N reason why the ordinary h, u restriction is tracial after disorder averaging. For any word $F(J)$ in jointly covariant operators independent of the fixed choice O, $\mathbb{E}F(J)$ commutes with the irreducible $U(N)$ action on $\bigwedge^m \mathbb{C}^N$ and is scalar. Thus $\mathbb{E}\langle O|F(J)|O\rangle = \mathbb{E}\,\mathrm{tr}_m\, F(J)$. The selected M depends on O and does not satisfy this argument.

7 A stationary state retaining noncommutative dynamics

By the all-word theorem, $\varphi|_{\mathcal{D}}$ is the ordinary two-color q-Gaussian trace: pair equal colors with crossing weight q. Write $\alpha_t(A) = e^{ith}Ae^{-ith}$. Pure absolute continuity of h implies $e^{-ith}\Omega \rightharpoonup 0$ as $|t| \to \infty$, so compact expectations vanish. Equation (17) gives

$$\lim_{|t|\to\infty} \varphi(\alpha_t(B+K)) = \varphi(B) =: \tau_\infty(B+K), \qquad B \in \mathcal{D}, \quad K \in \mathcal{K}(\mathcal{H}). \tag{25}$$

The limit of states proves that this expression is well defined, positive and normalized, including when the decomposition is not unique. Products with a compact factor are compact; the trace on $\mathcal{D}$ therefore makes τ_∞ tracial on all of $\mathcal{A}_+$. It is stationary and annihilates the compact ideal.

In this particular state's GNS representation, compact operators vanish and the h, u moments are the standard q-Gaussian moments. The generated von Neumann algebra is consequently $\Gamma_q(\mathbb{R}^2)$, a Type II_1 factor by Ricard's theorem [4]. This identifies the GNS algebra of τ_∞; it neither identifies the full C^* quotient $\mathcal{A}_+/\mathcal{K}(\mathcal{H})$ nor classifies its other states.

The surviving channel has nontrivial h dynamics. Wick evaluation gives $\tau_\infty(hu^2h) = \tau_\infty(uh^2u) = 1$ and $\tau_\infty(huhu) = \tau_\infty(uhuh) = q$, hence

$$\tau_\infty([h,u]^\dagger[h,u]) = 2(1-q) > 0. \tag{26}$$

Thus the previous stationary-state obstruction has a constructive resolution after adding a specified microscopic observable. It still applies to the smaller norm algebra. The new stationary state is singular relative to the original Type I representation, since it kills all compacts. It is not a trace-class density on $B(\mathcal{H})$.

For every finite real β, boundedness of h allows

$$\tau_\beta(A) = \frac{\tau_\infty(e^{-\beta h}A)}{Z_\beta}, \qquad Z_\beta = \tau_\infty(e^{-\beta h}) > 0. \tag{27}$$

Tracial cyclicity proves the KMS identity $\tau_\beta(A\alpha_{i\beta}(B)) = \tau_\beta(BA)$. These are genuine KMS states for the specified inner flow on $\mathcal{A}_+$. Assigning β to a de Sitter temperature requires an additional physical dictionary. There is no contradiction with the absence of normal equilibrium in the original full $B(\mathcal{H})$, nor with the previous absence of finite-temperature KMS states on that full algebra. We changed the norm algebra and its state representation.

8 The surviving spectral correlation and numerical evidence

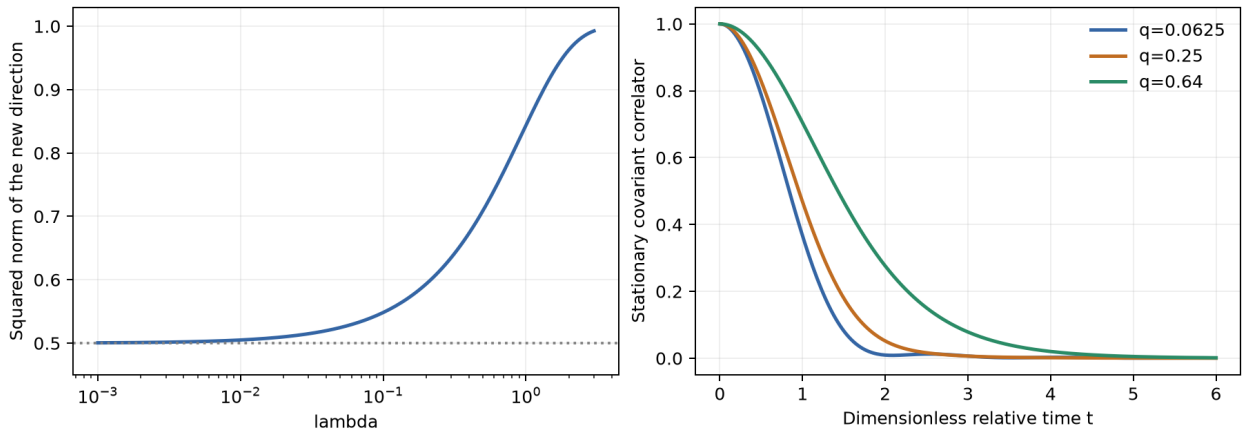
Let μ_q be the normalized vacuum spectral measure of the Jacobi matrix with off-diagonals $\sqrt{[n+1]_q}$, and p_n its orthonormal polynomials, $p_0 = 1$. Its support is $[-E_*, E_*]$, $E_* = 2/\sqrt{1-q}$. The ordinary matter insertion between two energy chords has the standard q-Mehler kernel

$$K_q(E, E') = \sum_{n \geq 0} q^n p_n(E) p_n(E'). \tag{28}$$

Applying that existing formula to our surviving field gives

$$\begin{aligned} C_u(t) &= \tau_\infty(u \alpha_t(u)) && (29)\\ &= \iint e^{it(E'-E)} K_q(E, E')\, \mathrm{d}\mu_q(E) \mathrm{d}\mu_q(E') && (30)\\ &= \sum_{n \geq 0} q^n \left| \langle n | e^{ith} | 0 \rangle \right|^2 = 1 - (1-q)t^2 + O(t^4). && (31) \end{aligned}$$

This is an application of the known kernel, not a new general two-energy transform. For a Jacobi cutoff of dimension L, bandedness makes the first L Taylor vector coefficients exact. With $x = E_*|t|$, an exact-arithmetic correlation error bound is $4e^x x^L/L!$. It does not enclose floating-point roundoff.



The left panel plots (20); the right panel plots the stationary correlation. The physical examples use $q = 0.0625, 0.25, 0.64$ and $\lambda = -\log q/4$. The 18 saved time rows use $t = 0, 0.5, 1, 2, 4, 6$, with Jacobi dimensions 80 and 120, and include finite-difference curvature diagnostics. At $t = 0$ the truncation error is exactly zero; its logarithmic bound is stored as null.

9 Reproducibility, scope and the next microscopic task

Three successful scientific runs were sealed before writing this manuscript. No successful driver was rerun for presentation.

Output	Independent checks
Site	136704 exact CAR matrix comparisons at $(N,p) = (4,1),(4,2),(6,1)$; 7880 double-cluster coefficient checks through three edges.
Words	8192 sixth words; 1458 old-restriction comparisons; 720 algebra relations; 62 projection orthogonalities; 1638 Gram entries through level three.
Spectrum	Three physical q values, 18 time rows, 80/120 Jacobi comparison, analytic cutoff bounds and curvature diagnostics.

The exact word tests use rational $(\gamma, d) = (1/2, 1/2), (2/3, 1/2)$ as independent algebra parameters, with $c_U = -d\gamma$. They are not points on the physical curve $q = e^{-4d^2}$. Positive pivots in those finite Gram tests do not establish the infinite positive representation; positivity above comes from the microscopic limiting moments. The site checker independently applies

$$2(p - \mathcal{L})X = (2p - N)X + \sum_i (c_i^\dagger X c_i + c_i X c_i^\dagger) \tag{32}$$

and compares it with the marked-site construction. Numerical versions are Python 3.14, NumPy 2.4.6, SciPy 1.18.0 and Matplotlib 3.11.0. The stage manifest binds inputs, parameters, outputs, build logs and page renders by SHA256. Floating diagnostics are not interval proofs.

The next task is the actual unpinched neutral-slope rank symbol and selected-state control of pinching, followed by J_2 and the complete bilocal in the common representation. The previously completed full-trace bilocal four-point construction is an input, not an open task to repeat. The spatial and mass generators, the de Sitter temperature assignment and finite-G dynamics remain to be derived.

References

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[5] Xu, arXiv:2403.09021, version 3, Sec. 3.3, Eq. (75).